IMPORT–EXPORT B2B E-COMMERCE PLATFORM

Software Requirements Specification (SRS) &
Functional Requirements Document (FRD)

Comprehensive requirements for marketplace, lot-wise pricing, enquiries, orders, trade documents, payments, shipments, containers, roles, reporting and Tally integration.

Document Control	Details
Document type	Business requirements and functional specification
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Status	Draft for business validation and development planning
Intended audience	Business owners, product manager, UI/UX, frontend, backend, QA, DevOps and integration teams

Purpose: Define what the platform must do, the data it must store, who can use each feature, how each workflow progresses, and the acceptance criteria for delivery.

Confidential business planning document. Product, trade, tax, customs, banking and document requirements must be validated for the actual goods, countries, routes and applicable regulations before production use.

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1. Executive Summary and Objectives

The proposed solution is a role-based B2B import–export trade and logistics platform. It combines product discovery and enquiries with commercial negotiation, documentation, payments, shipment visibility and historical order management.

1.1 Business objectives

- Publish products by category, specification, grade and packaging type for domestic and international buyers.
- Maintain multiple lots for one product, each with its own upload timestamp, price, quality, supplier and available quantity.
- Collect structured buyer enquiries with quantity, quality, destination, packaging, delivery needs and Incoterms.
- Generate and manage quotations, proforma invoices, agreements, packing lists, tax/commercial invoices and trade certificates.
- Track order milestones, payment milestones, documents, vehicles, shipments, ports, containers and estimated/actual arrival dates.
- Provide distinct workspaces for Super Admin, Sub Admin, Developer Admin, Supplier, Importer/Buyer, CHA and domestic/international users.
- Integrate accounting with Tally and optionally integrate payment gateways, carrier tracking, GPS, messaging and e-signature providers.
- Maintain a complete, auditable record of changes, approvals, payments, files and shipment events.

1.2 Success outcomes

- Users can determine which product lot is available and at what currently approved price.
- Buyers can submit enquiries and follow them through quotation, agreement, payment, order and delivery.
- Operations staff can identify missing documents, outstanding payments, delayed shipments and clearance blockers.
- Business records are consistent across product inventory, orders, invoices, shipment records and accounting integrations.

2. Scope and Assumptions

2.1 In scope

- Responsive public marketplace and authenticated portals.
- Role-based user and company management.
- Categories, products, lots, pricing, stock and price history.
- RFQs/enquiries, negotiation, quotation, agreement, order and past-order history.
- Payment recording and verification, invoice generation and outstanding balance tracking.
- Document upload, generation, approval, versioning, expiry and secure download.
- Shipment, vehicle, port, container and tracking-event management.
- Supplier and CHA workflows, notifications, dashboards and reports.
- Tally integration design and operational monitoring.
- Security, audit logs, backup, testing and production-readiness requirements.

2.2 Out of scope unless separately approved

- Automatic legal or customs approval by the platform.
- Guaranteed live GPS or vessel/container location without a connected provider or verified event source.
- Automatic foreign-exchange settlement or banking services without a suitable provider and business/legal review.
- Automatic legal validity of a typed name or uploaded signature image.
- Unverified claims about regulatory compliance, tax calculation, certificate issuance or customs clearance.

2.3 Working assumptions

- The platform may support both domestic and international trade; business rules can vary by transaction type.
- An order may have multiple order items, lot allocations, shipments, containers, payments and documents.
- Supplier product publishing and special pricing may require administrator approval.
- External integrations are provider-dependent and must be confirmed before implementation.
- All dates are stored consistently and displayed in the appropriate local time zone.
- Currency, unit of measure, tax treatment and Incoterms are explicit fields, not free-text assumptions.

3. User Roles and Permissions

Role	Core responsibilities	Access boundaries
Super Admin	Platform-wide settings, users, roles, approvals, products, orders, reports, integrations and audit logs.	Full authorized business access; sensitive actions must still be logged.
Sub Admin	Assigned product, order, documentation, supplier, buyer or shipment operations.	Restricted to assigned modules, records and actions.
Developer Admin	Technical configuration, API/webhook setup, diagnostics, deployment and integration health.	Business data access minimized; production actions audited.
Supplier	Company profile, product lots, pricing, inventory, RFQ responses, order preparation and document submission.	Own company's products, assigned enquiries/orders and authorized records.
Importer / Buyer	Browse products, submit RFQs, negotiate, accept quotations, sign agreements, pay and track orders.	Own company's enquiries, quotations, orders, payments and shared documents.
CHA (Customs House Agent)	Assigned customs tasks, document submission, filing references, clearance updates and remarks.	Only assigned orders/shipments and relevant documents.
Domestic User	Domestic catalogue, enquiry, quotation, order, payment and delivery tracking.	Domestic transactions permitted for the account.
International User	International catalogue, currency/Incoterm details, export enquiries, documents and shipment tracking.	International transactions permitted for the account.

3.1 Common account capabilities

- Registration/invitation, login, logout, password reset, account activation and session management.
- Company profile, contacts, billing/shipping addresses and relevant tax/trade information.
- Role-based menus and permissions for view, create, edit, approve, download, export, cancel and delete.
- Notifications, activity history, user preferences and support/contact information.
- Optional MFA for privileged users and configurable session expiry.

3.2 Permission matrix

Module	Super Admin	Sub Admin	Supplier	Buyer	CHA	Developer Admin
Users & roles	Manage all	Assigned scope	Own users if allowed	Own users if allowed	No	Technical configuration
Products/lots	Full	Assigned scope	Own products/lots	Read published data	No	No routine edits
RFQs/quotations	Full	Assigned scope	Assigned RFQs	Own RFQs	No	No routine edits
Orders	Full	Assigned scope	Assigned orders	Own orders	Assigned shipment context	Diagnostics only
Payments	Full/authorized	Assigned scope	Authorized summaries	Own payments	No	No routine access
Documents	Full/authorized	Assigned scope	Own/assigned	Own/shared	Assigned documents	System diagnostics
Shipment/clearance	Full	Assigned scope	Assigned shipments	Own shipment view	Assigned clearance	Integration diagnostics
Tally/system settings	Authorized	Limited	No	No	No	Technical integration access

Permission names and scopes must be configurable. The server must enforce permissions; hiding a menu item in the UI is not sufficient security.

4. Product Catalogue, Categories and Lot-wise Pricing

4.1 Category fields

- Category ID, category name, parent category ID, description, image, display order, active/inactive status, created by, created date and updated date.

4.2 Product fields

Field name	Requirement
product_id / SKU	Unique stable product identifier and human-readable SKU.
product_name	Required display name.
category_id / subcategory_id	Category hierarchy.
description	Product description and usage.
specifications	Structured technical/quality attributes.
grade / quality	Commercial grade or quality standard.
unit_of_measure	Kg, metric tonne, piece, box, litre or configured unit.
packaging_type	Bag, carton, pallet, bulk or configured packaging.
origin_country	Country of origin where applicable.
minimum_order_quantity	Minimum requestable quantity and unit.
images / attachments	Approved product images and specifications.
domestic_visibility / export_visibility	Where the product is available.
publication_status	Draft, pending approval, active, inactive or archived.
created_by / created_at / updated_at	Audit and maintenance fields.

4.3 Product lot fields

Field name	Requirement
lot_id / lot_code	Unique identifier for each stock lot.
product_id / supplier_id	References to the product and responsible supplier.
uploaded_at / effective_at	Server timestamp for upload and price effectiveness.
warehouse_location	Warehouse/site and stock location.
quality_grade / inspection_status	Lot-specific quality and inspection details.
quantity_received	Original received quantity and unit.
quantity_reserved	Quantity reserved for orders.
quantity_dispatched	Quantity shipped/dispatched.
quantity_adjusted	Audited adjustment quantity and reason.
quantity_available	Calculated or transactionally maintained available quantity.
unit_price / currency	Lot-specific commercial price and currency.
domestic_price / export_price	Optional separate price schedules.
incoterm / named_place	Applicable commercial delivery term and named place/port.
packaging_details	Lot-specific packaging and weights.
manufacture_or_harvest_date	Optional product-specific date.
expiry_date	Optional expiry/best-before date where relevant.
certificates / photos	Lot-level evidence and supporting files.
lot_status	Draft, pending approval, active, reserved, depleted, blocked or archived.

4.4 Lot pricing rules

- Every lot has its own price, currency, unit, upload timestamp and effective timestamp.
- Uploading a new lot must not overwrite another lot's price, stock, quality or history.
- Pricing can be manual, governed by date-based rules, or a combination, as selected by the business.
- Date-based rules must define their formula, timezone, rounding, currency, effective period, priority and approval owner.
- Price changes create a price-history record with old price, new price, actor, reason, effective time and approval status.
- Existing accepted quotations and confirmed orders preserve their agreed price snapshots.
- Bulk/quantity pricing, minimum order quantities, negotiated discounts and quotation expiry can be configured.
- Domestic and export price lists may differ; freight, insurance, taxes and other charges must be represented separately where required.
- Do not assume a newer lot is more expensive; apply only the configured pricing policy.

Example: Product PROD-001 may have Lot A at ■48,000 per tonne with 20 tonnes available and Lot B at ■51,000 per tonne with 35 tonnes available. These are illustrative values. A 30-tonne order may allocate stock from both lots if quality and buyer terms permit.

5. Inventory and Stock Reservation

5.1 Inventory capabilities

- Receive stock into a lot and record source, date, quantity and evidence.
- Reserve stock when a quotation/order reaches the configured commitment point.
- Release reservations on expiry, rejection, cancellation or authorized amendment.
- Record dispatch, delivery, returns, damage, loss and manual adjustments.

- Support partial fulfilment and multiple lots per order item.
- Show on-hand, reserved, dispatched, delivered and available quantities separately.
- Maintain stock movement ledger with actor, reason, timestamp and reference.
- Support low-stock alerts, lot ageing and optional warehouse transfers.

5.2 Inventory formula

Available quantity = received/on-hand quantity + accepted returns – reserved quantity – dispatched quantity – blocked quantity, subject to the exact inventory model chosen. The implementation must avoid subtracting dispatched quantities twice if dispatch already reduces on-hand stock.

5.3 Concurrency and allocation

- Use database transactions/locking or equivalent concurrency control for reservations.
- Validate units and convert quantities only through configured conversion rules.
- Prevent negative stock unless a privileged override is explicitly enabled and audited.
- Store lot allocation per order item and preserve the original allocation history.
- Allow authorized substitution/reallocation only with a recorded reason and buyer approval when commercial terms change.

6. Buyer Marketplace and Enquiry / RFQ

6.1 Marketplace screens

- Home and featured products; category and subcategory pages; search; filters; product details; lot selector; company/profile pages where enabled; RFQ form; enquiry history; quotation comparison; order tracking; and document downloads.

6.2 Search and filters

- Product name/SKU, category, quality/grade, origin, supplier, price range, quantity available, unit, domestic/export availability, destination eligibility and active lot status.

6.3 RFQ fields

Field	Details
rfq_id / buyer_id	Unique enquiry and buyer/company reference.
product_id / preferred_lot_ids	Requested product and optional lot preferences.
requested_quantity / unit	Requested quantity and measurement unit.
quality_specification	Grade, tolerance, inspection and technical requirements.
packaging_requirement	Pack type, size, palletization and labelling requirements.
trade_type	Domestic or international.
destination	Country, city, port, warehouse or delivery address.
requested_delivery_window	Preferred date or range.
incoterm / named_place	Requested Incoterm and named port/place.
currency	Requested quotation currency.
required_documents	Buyer’s requested certificates/documents.
notes / attachments	Additional requirements and permitted files.
status / assigned_to	Workflow state and responsible sales/supplier user.
created_at / updated_at	Audit timestamps.

6.4 RFQ workflow

Draft → Submitted → Under Review → Clarification Needed / Supplier Responding → Quotation Issued → Accepted / Rejected / Expired → Converted to Order or Closed.

- Validate required fields and quantity/unit.
- Show stock availability as a current estimate and revalidate before reservation.
- Allow clarification messages and attachments.
- Support one RFQ receiving one or more supplier offers if the business uses multi-supplier sourcing.
- Keep buyer-visible communications separate from internal notes.
- Record the reason for rejection, expiry or closure where applicable.

7. Chatbot-assisted Ordering

7.1 Guided question sequence

- Identify the product/category.
- Ask requested quantity and unit.
- Collect quality grade and specifications.
- Collect packaging and labelling needs.
- Ask domestic/international and destination country/port/address.
- Ask requested delivery date/window.
- Collect preferred Incoterm and currency.
- Show matching products/lots using verified inventory and price data.
- Summarize the request and ask the user to confirm before creating the RFQ.
- Save the RFQ against the authenticated user and show the reference number.

7.2 Chatbot controls

- Use backend APIs as the source of truth for prices, lots and stock.
- Clearly distinguish indicative prices from approved quotations.
- Never claim an order, payment, agreement or shipment is confirmed without a successful backend state change.
- Ask for missing required details; allow saving a draft if the user is not ready.
- Escalate negotiation, unclear requirements, unusual quantities and compliance questions to an authorized human.
- Log AI-generated summaries and user confirmation; allow correction before submission.
- Protect personal/company data and restrict retrieval to the user’s authorized records.

8. Quotations, Proforma Invoices and Agreements

8.1 Quotation / estimate

- Quotation number; issue date; expiry date; buyer/seller details; product and lot allocation; quantity/unit; unit price; currency; discounts; tax; freight; insurance; other charges; Incoterm/named place; delivery window; payment terms; validity; exclusions; notes; preparer; approval status; and version.

8.2 Proforma invoice (PI)

- PI number; linked RFQ/quotation/order; seller and buyer legal details; product descriptions; lot/quantity; unit price; line totals; currency; charges; tax where applicable; total; payment instructions/terms; delivery/shipping terms; bank details as approved; validity; signature/approval; issue and revision history.

8.3 Agreement / contract

- Contract ID; parties; authorized signers; referenced quotation/PI; goods and specifications; quantity tolerance; price/currency; payment schedule; delivery obligations; Incoterm version and named place; inspection/quality provisions; packaging; documentation responsibilities; shipment schedule; claims/cancellation terms; governing-law/dispute clauses supplied by legal counsel; signatures; signed timestamps; and version history.

8.4 Document workflow

Draft → Internal Review → Approved for Issue → Sent to Counterparty → Accepted / Rejected / Expired → Superseded or Cancelled where appropriate.

- Generate PDFs from saved order and company data using versioned templates.
- Prevent unapproved drafts from being represented as issued documents.
- Record acceptance method, actor and timestamp.
- Changes after acceptance require a controlled amendment/revision process.
- Document numbering and numbering resets must be configurable and auditable.

9. Order Management and Past Orders

9.1 Order record fields

- order_id; order_number; buyer_id; supplier_id; source_rfq_id; accepted_quotation_id; agreement_id; order_date; order_type; status; currency; Incoterm and named place; total quantity; subtotal; discounts; taxes; freight; insurance; total amount; amount paid; balance due; expected delivery; assigned operations user; and audit timestamps.

9.2 Order items

- order_item_id; product_id; selected lot allocation(s); agreed product description/specification snapshot; quantity and unit; unit price; currency; discount; taxes/charges; line total; accepted quality; packaging; and allocated/dispatched/delivered/outstanding quantities.

9.3 Order lifecycle

Stage	Meaning / exit condition
Enquiry received	RFQ exists and is assigned for review.
Quotation issued	Approved quotation has been sent.
Agreement pending	Contract or required terms await acceptance/signature.
Order confirmed	Required acceptance and internal checks have passed.
Payment pending / partial	Required payment remains outstanding.
Goods preparation	Stock, quality checks and packaging are being completed.
Ready for dispatch	Goods and required pre-dispatch checks are complete.
Dispatch / export processing	Transport, customs and documents are in progress.
Loaded / departed	Relevant vehicle/vessel departure evidence is recorded.
In transit	Shipment is moving; latest verified event is available.
Arrived / clearance	Destination arrival and clearance steps are tracked.
Delivered / completed	Delivery evidence and configured closing checks are complete.
On hold / cancelled / disputed	Exception states with reason, actor and resolution history.

9.4 Past orders

- Filter by order date, product, supplier, buyer, country, destination, order status, payment status, shipment status, Incoterm and order value.
- Display the final agreed prices and lot allocation, not current catalogue prices.

- Show document history, payment history, shipment events, chat/notes, amendments, cancellations, disputes and proof of delivery.
- Support reorder or copy-as-new-RFQ without automatically reusing expired prices or old stock allocations.

9.5 Multiple shipments

An order may be fulfilled through one or more shipments. Each shipment must have its own quantity allocation, route, carrier/container references, documents, milestone history and delivery status. The order-level status should be derived from or reconciled with all its shipment and quantity states.

10. Payments and Financial Records

10.1 Payment types

- Advance payment; deposit; partial payment; balance payment; full payment; freight/insurance charge; refund; credit note; adjustment; and other configurable payment categories.

10.2 Payment fields

- payment_id; buyer/payer; payee; order_id; invoice_id; amount; currency; payment method; transaction/reference number; initiated_at; received_at; due_date; proof attachment; verification source; verified_by; status; refund/adjustment link; gateway response reference; and audit timestamps.

10.3 Payment statuses

Draft → Pending → Initiated → Verification Required / Processing → Partially Paid / Paid, or Failed / Rejected / Cancelled / Refunded / Disputed. Use separate status and amount fields where necessary to avoid ambiguous states.

10.4 Financial rules

- Maintain invoice total, verified paid amount, credits/refunds and balance due separately.
- Payment proof upload does not automatically mean funds have been received.
- Use payment-provider confirmation or authorized finance verification.
- Support partial payments and multiple receipts against an invoice.
- Reconcile currency and payment references; retain exchange-rate source/date when conversion is used.
- Do not store raw card credentials; use a compliant payment provider.
- Record due dates, overdue state, reminders, reconciliation and manual adjustments.
- Apply role permissions to bank details, payment records and financial exports.

11. Trade Documentation Management

Document name	Purpose / key fields	Workflow / notes
Quotation / Estimate	Prices, quantities, validity, delivery/payment terms.	Draft, review, issue, accept/reject, expire.
Proforma Invoice (PI)	Preliminary transaction value, goods, terms and payment instructions.	Generated from approved commercial data; versioned.
Packing List	Package marks/numbers, package count, contents, net/gross weights, dimensions and units.	Prepared, checked, finalized; linked to shipment.
Tax Invoice	Applicable seller/buyer details, invoice number/date, goods/services, taxable value, tax and totals.	Rules depend on jurisdiction and transaction type.
Commercial Invoice	Goods description, quantity, value, currency, buyer/seller and trade terms.	Use applicable template; avoid confusing it with domestic tax invoice.
Phytosanitary Certificate	Plant/plant-product health certification where applicable.	Issued by competent authority; platform stores evidence.
Fumigation Certificate	Treatment details, date, provider and relevant goods/container.	Required only where applicable.
Certificate of Origin (COO / C/O)	Goods origin, exporter/importer and issuing authority details.	Store issuer, certificate number, issue date and file.
CHA / Customs Documents	Filing references, declarations, examination and release evidence.	Document set varies by country, commodity and procedure.
Bill of Lading (B/L)	Carrier, shipper, consignee, vessel/voyage, ports, goods, packages and issue details.	Draft/review/issued/amended; retain final versions.
Inspection / Quality Certificate	Inspector, standard, results, lot and inspection date.	Optional/configurable by commodity and contract.
Insurance Certificate / Policy	Insured party, coverage, shipment, insured value and validity.	Where required by agreed terms.
Transport / Delivery Proof	Vehicle/waybill/delivery note, recipient and receipt evidence.	Applicable for road/rail/air/final-mile transport.
Other configurable document	Any additional document type defined by business/admin.	Custom metadata, checklist and approval rules.

11.1 Document metadata

- document_id; document_type; order_id; shipment_id; lot_id if applicable; uploader; issuer; document_number; issue_date; expiry_date; file_storage_key; file_hash; version; parent/superseded_document_id; status; review comments; approved/rejected_by; timestamps; and access classification.

11.2 Document status list

Missing → Draft → Uploaded / Submitted → Under Review → Approved / Rejected → Issued / Finalized → Expired / Superseded / Cancelled where applicable.

11.3 Document management requirements

- Allow PDF/JPG/PNG and other approved formats; enforce file size and MIME/type checks.
- Store files in private object/file storage; use authorized, time-limited downloads.
- Support version history, replacement reason, checksum and audit trail.
- Allow document checklist templates by transaction type, commodity, country, route and Incoterm.
- Show required/optional, pending, rejected, expired and approved counts.
- Prevent a draft or rejected document from satisfying a mandatory final-document requirement.
- Do not claim that a document was officially issued by an authority merely because a user uploaded it.

The terms “Photo sanitary/Sito sanitory” are interpreted as “Phytosanitary Certificate.” Confirm the intended certificate with the business. Certificate, tax, customs and shipping-document requirements must be validated for the actual product and destination.

12. Shipment, Vehicle, Container and Port Tracking

12.1 Shipment fields

- shipment_id; order_id; shipment_number; mode; origin address/warehouse; origin port; destination port; final destination; carrier; freight forwarder; booking reference; tracking reference; vessel name; voyage number; departure date; estimated arrival; actual arrival; assigned CHA; status; and timestamps.

12.2 Container fields

- container_id; container_number; container_type/size; seal_number; shipment_id; loaded quantity/units; gross/net weight; assigned date; current reported location; last event time; carrier tracking reference; status; and assignment history.

12.3 Vehicle and trip fields

- vehicle_id; registration number; vehicle type; carrier/owner; driver/contact where authorized; trip_id; shipment/order reference; pickup/delivery locations; dispatch time; GPS provider/device reference; last location/time; trip status; and proof of delivery.

12.4 Tracking event fields

- event_id; shipment_id; container_id if applicable; event_type; event_status; location name; coordinates only if provided by a trusted source; planned_at; actual_at; source/provider; source_event_id; description; evidence attachment; created_by; and recorded_at.

12.5 Shipment milestones

Milestone	Fields to capture
Order received	Order reference, confirmation timestamp.
Payment received	Verified amount, currency, receipt reference and timestamp.
Goods ready	Warehouse, quantity, inspection/packing completion and ready date.
Vehicle assigned / pickup	Vehicle/carrier reference, pickup location, driver where permitted, planned/actual time.
Origin port reached	Port/location, arrival time, supporting event/evidence.
Customs processing	Filing reference, status, query/hold, responsible CHA, required document and timestamps.
Loaded onto vessel	Vessel, voyage, port, loading date/time and available B/L reference.
Vessel departed	Departure port and actual departure timestamp.
In transit / transshipment	Latest source event, location, event time and revised ETA.
Destination port reached	Port, actual arrival, terminal/carrier event.
Customs cleared / released	Release status, reference, actual date and evidence.
Out for delivery	Last-mile carrier, vehicle/tracking reference and ETA.
Delivered	Delivered quantity, date/time, recipient/proof of delivery and exceptions.

12.6 Tracking source and ETA rules

- Distinguish manual updates, carrier API events, GPS-derived locations and estimated events.
- Every location update must include source and event timestamp; show the last update time and timezone.
- Do not fabricate a container’s live location if no reliable tracking feed exists.
- Keep historical ETAs when the estimate changes and record the source/reason.
- Support delayed, held, rerouted, transshipment, damaged, lost and exception states.
- Show vehicle GPS only when a compatible device/provider is connected and permitted.
- Support manual event entry for MVP, with clear “manually updated” labelling.

12.7 FOB and CIF

- Store Incoterms rule, version and named port/place on the accepted commercial record.

- FOB: record the agreed vessel-loading milestone at the named port and evidence where available.
- CIF: record the named destination port and the seller-arranged cost, insurance and freight details as applicable to the contract.
- Track estimated arrival separately from actual arrival and contractual delivery obligations.
- Do not infer legal transfer of risk solely from a generic app status; apply the agreed Incoterms rule and legal/commercial review.

13. Domestic and International Workflows

Area	Domestic workflow	International workflow
User access	Domestic account permissions and local transaction rules.	International account permissions and trade-related verification.
Price/currency	Configured domestic price and currency, typically INR where applicable.	Agreed transaction currency and export price schedule.
Destination	Domestic address, warehouse or delivery location.	Destination country, port, terminal and final delivery location.
Taxes/documents	Applicable domestic invoice, tax and transport records.	Applicable export/import, customs, origin, shipping and certificate requirements.
Transport	Road, rail, air or other supported local mode.	Sea, air, road, rail or multimodal shipment.
Tracking	Pickup, dispatch, in-transit and delivery milestones.	Origin port, customs, vessel/container, transshipment, destination port and final delivery.
Payment	Configured local payment and reconciliation methods.	Agreed international payment/collection process and currency records.

Use a shared order engine with configurable domestic/international rules rather than maintaining two unrelated systems.

14. Supplier Portal

- Company onboarding, contacts, addresses and verification status.
- Product/category selection, SKU, specifications, images and document upload.
- Lot creation, upload timestamp, quantity, price, currency, quality and packaging.
- Inventory movement, reservations, dispatch and stock alerts.
- RFQ inbox, clarification, quotation drafting and negotiation.
- Order acceptance, production/preparation, inspection and ready-to-dispatch status.
- Document submission and rejection/approval feedback.
- Assigned shipment visibility and milestone updates where permitted.
- Relevant payment/receivable summaries and invoice history.
- CSV/Excel bulk import with validation preview, row-level errors and safe rollback.

15. CHA Portal

- Assigned order/shipment queue with deadlines and priority.
- Customs checklist based on route, commodity and transaction configuration.
- Document upload, review state, comments and resubmission.
- Customs filing/declaration reference numbers and submission timestamps.
- Examination, query, hold, duty/charge information where authorized, clearance and release status.
- Notification of missing/rejected documents and approaching deadlines.
- Communication with supplier/importer/operations users for assigned records.
- Downloadable, access-controlled document package.
- Audit history of every status update and attachment.

- Permission restrictions against unrelated pricing, payments, products and orders.

16. Tally Integration

16.1 Candidate synchronized data

- Customer and supplier ledgers; sales invoices; credit notes; purchase records if used; receipts; payments; stock items and units; tax/ledger mappings; currency and accounting references.

16.2 Integration requirements

- Confirm Tally product/version, hosting environment, connectivity and authorized integration method.
- Map platform entities to Tally ledgers, stock items, vouchers, tax ledgers and units.
- Define source of truth for invoice numbering, stock and accounting entries.
- Use stable external IDs and idempotency keys to prevent duplicate voucher creation.
- Maintain sync queue, status, attempt count, timestamps, payload reference, response/error and reconciliation result.
- Support retry with safe handling, manual re-sync by authorized users and exception reporting.
- Do not log passwords, tokens or unnecessary sensitive financial data.
- Test create/update/cancel/amend scenarios in a non-production environment before live sync.

16.3 Suggested sync statuses

Not Queued → Queued → Processing → Synced, or Failed → Retry Scheduled / Manual Review. Reconciliation may be a separate state so that a successful API call is not mistaken for a fully reconciled account.

17. Dashboards, Reports and Exports

Dashboard	Key information
Super Admin	Products/lots, RFQs, quotations, confirmed orders, order value, verified receipts, outstanding balances, pending documents, shipment delays, supplier/buyer activity, integration failures and audit alerts.
Supplier	Own products/lots, available/reserved stock, enquiries, quotation tasks, orders to prepare, required documents and authorized receivable summaries.
Buyer/Importer	Recent enquiries, quotations awaiting action, agreements to sign, payments due, active orders, shipment ETAs, documents and past orders.
CHA	Assigned shipments, documents pending/rejected, customs queries/holds, deadlines, clearance status and completed records.
Sub Admin	Tasks and KPIs limited to assigned operational scope.
Developer Admin	API health, job queues, error rates, integration failures, deployment/system status and technical audit events.

17.1 Report catalogue

- Sales by product/category/supplier; lot-wise price history; inventory and stock ageing; low-stock report; RFQ conversion; quotation win rate; order status; outstanding receivables; payment reconciliation; document completeness/expiry; shipment delay; delivery performance; container movement; CHA clearance workload; and Tally sync failures.

17.2 Export controls

- Support filtering, sorting, pagination, CSV/Excel and PDF where useful.
- Apply the same authorization to exported data as to on-screen data.
- Record sensitive exports and include generated-at time, timezone and filter parameters.
- Large exports should run asynchronously with status and secure download expiry.

18. Notifications, Chat and Activity History

18.1 Notification events

- New RFQ; supplier response; quotation issued/accepted/rejected/expiring; agreement signature request; order confirmed/amended/cancelled; payment due/received/failed/overdue; document uploaded/rejected/approved/expiring; goods ready; vessel departed; container milestone updated; ETA changed; shipment delayed; customs hold/release; delivery completed; and Tally sync failed.

18.2 Delivery channels

- In-app notifications and email are baseline options. SMS/WhatsApp can be added through approved providers, templates, opt-in and applicable messaging rules.

18.3 Order chat

- Order-linked conversations with authorized participants; message text; attachments; sender; sent/edited timestamps; read status if supported; moderation/reporting controls; and retention policy.
- Separate external participant chat from internal staff notes.
- Ensure documents and chat attachments inherit appropriate order permissions.

19. Administration and Audit Logs

19.1 Administration

- Manage user invitations, account status, role/permission templates, categories, units, currencies, Incoterms versions, document types, checklist templates, numbering rules, notification templates, pricing policies and integration configuration.

19.2 Audit log fields

- audit_id; actor_user_id; actor_role; action; entity_type; entity_id; timestamp; source/IP/device metadata where appropriate; before/after diff or safe change summary; reason; request/correlation ID; and result.

19.3 Audit requirements

- Log price changes, approvals, payment verification, document changes, order transitions, permissions, shipment events, inventory adjustments and Tally sync actions.
- Prevent ordinary users from modifying or deleting audit records.
- Avoid placing passwords, access tokens, full payment credentials or unnecessary personal data in logs.

20. Data Model and Entity Register

Entity/table	Key fields / relationships
Users	user_id, company_id, role_id, name, email/phone, status, authentication references, timestamps.
Roles / Permissions	role_id, permission_key, action, scope, assignment and policy metadata.
Companies	company_id, legal/trading name, country, registration/tax references, contacts, addresses, verification status.
Categories	category_id, parent_id, name, description, image, status, display order.
Products	product_id, SKU, category_id, name, description, specifications, unit, status, visibility.
ProductLots	lot_id, product_id, supplier_id, location, uploaded_at, effective_at, quality, quantity, price, currency, status.
PriceHistory	price_history_id, lot_id, old/new price, currency, reason, actor, effective_at, approval.
InventoryMovements	movement_id, lot_id, movement type, quantity/unit, reference, reason, actor, timestamp.
RFQs / Enquiries	rfq_id, buyer_id, product references, quantity, requirements, destination, terms, status, assignment.
Quotations / Items	quotation_id, rfq_id, version, validity, currency, totals, terms; items with product/lot/price/quantity.
Orders / OrderItems	order_id, buyer/supplier, accepted quote, status, totals, terms; items and immutable price/lot snapshots.
Agreements / Signatures	agreement_id, order_id, version, document reference, signers, status, signed timestamps.
Invoices	invoice_id, order_id, type, number, issue/due dates, currency, line totals, tax/charges, status.
Payments / Adjustments	payment_id, invoice/order, amount, currency, reference, status, verifier, refunds/credits and timestamps.
Documents	document_id, type, order/shipment/lot, storage key, version, status, issuer, issue/expiry, review history.
Shipments	shipment_id, order_id, mode, route, carrier, booking/tracking references, vessel, dates and status.
Containers	container_id, shipment_id, number, type, seal, quantity/weight, status and assignment history.
Vehicles / Trips	vehicle_id, registration, carrier, driver reference; trip_id, route, assigned shipment and trip status.
TrackingEvents	event_id, shipment/container/trip, event type, location, event time, source, ETA, evidence and actor.
Messages / Conversations	conversation_id, order_id, participants; message_id, sender, content, attachment, timestamps.
Notifications	notification_id, recipient, event, channel, delivery/read status, timestamps.
AuditLogs	actor, action, entity, timestamp, change metadata, reason and correlation ID.
IntegrationLogs	provider, entity reference, external ID, sync status, attempt, error, response reference and timestamps.

20.1 Data integrity

- Use primary keys, foreign keys, unique constraints, appropriate indexes and database transactions.
- Use decimal/fixed-precision numeric fields for prices and monetary values; do not use floating-point for money.
- Store currency and unit codes explicitly; retain original and converted amounts where conversion is needed.
- Store timestamps in a consistent standard such as UTC and display local time appropriately.
- Use soft-delete/archive states for commercially important records rather than destructive deletion.
- Define retention and deletion policies for personal data, documents and audit events.

21. API and Integration Requirements

The routes below are proposed API groups, not existing or verified endpoints. Exact methods, schemas, authentication, validation, status codes and examples must be specified in the API contract.

API group	Proposed route prefix	Core operations
Authentication	/api/v1/auth	Login, logout, refresh/session, password reset, verification.
Users and companies	/api/v1/users; /api/v1/companies	Profiles, invitations, role assignments, company verification.
Categories	/api/v1/categories	List/create/update/archive categories.
Products	/api/v1/products	Search, filter, details, create/update, publish and archive.
Lots / inventory	/api/v1/lots; /api/v1/inventory	Lot CRUD, price changes, stock movements, reservation and release.

API group	Proposed route prefix	Core operations
RFQs	/api/v1/enquiries	Create/update/submit RFQ, clarification, assignment and closure.
Quotations	/api/v1/quotations	Draft, approve, issue, revise, accept/reject and expire.
Orders	/api/v1/orders	Create/confirm, list/detail, amend, hold/cancel and history.
Agreements	/api/v1/agreements	Generate, review, send, sign-provider callbacks and version history.
Invoices	/api/v1/invoices	PI/tax/commercial invoice generation, issue, amend and download.
Payments	/api/v1/payments	Initiate/record, webhook verification, receipt, reconciliation and refund.
Documents	/api/v1/documents	Upload, secure download, review, approve/reject, version and expiry.
Shipments	/api/v1/shipments	Create shipment, assign quantity, milestones, ETA and delivery proof.
Containers	/api/v1/containers	Assign/unassign, details, tracking lookup and event history.
Vehicles	/api/v1/vehicles	Vehicle/trip assignment, authorized GPS/event updates.
Chat	/api/v1/chat	Conversations, messages, attachments and read state.
Notifications	/api/v1/notifications	List, read/acknowledge and delivery status.
Reports	/api/v1/reports	Authorized filters, export jobs and secure download.
Tally	/api/v1/integrations/tally	Connection health, sync jobs, retries, mapping and reconciliation.

21.1 API standards

- Version APIs; define request/response schemas and consistent error objects.
- Use server-side authorization, validation, pagination, filtering and sorting.
- Use idempotency keys for order/payment/integration operations where applicable.
- Verify payment-provider webhooks and protect against replay attacks.
- Use signed/validated callbacks for external providers and store correlation IDs.
- Return clear conflict responses for stale prices, unavailable stock, expired quotations and invalid status transitions.
- Document APIs using OpenAPI/Swagger and provide a non-production test environment.

22. Security, Privacy and Non-functional Requirements

22.1 Security

- TLS in transit; secure password hashing; protected sessions/tokens; MFA for privileged users where appropriate; rate limiting; input/output validation; least privilege; record-level authorization; private file storage; secure download URLs; malware scanning; secrets management; dependency updates; and secure deployment configuration.

22.2 Privacy and compliance

- Collect only required personal/company information; define purpose, access, retention and deletion rules.
- Confirm applicable data-protection, tax, customs, export/import, electronic-signature and financial rules with qualified advisors.
- Restrict personal contact, banking and identity documents to authorized roles.
- Define audit retention and incident response procedures.

22.3 Reliability and recovery

- Automated database/file backups and restoration tests.
- Monitoring, error tracking, alerting, retry queues and dead-letter handling for failed integrations.
- Documented incident response, recovery objectives and escalation ownership.
- No critical financial/order operation should silently fail or partially update related records.

22.4 Performance and usability

- Responsive desktop/tablet/mobile layouts.
- Search, filters and pagination for large lists.
- Loading, empty, error, retry and permission-denied states on each screen.
- Accessible labels, keyboard support and readable contrast where applicable.
- Configurable performance, uptime, backup and recovery targets to be agreed before production.

22.5 Quality assurance

- Unit tests for pricing, stock, currency and status transitions.
- Integration tests for order/payment/document/shipment/Tally workflows.
- Permission tests for every role and record boundary.
- Concurrency tests for inventory reservation.
- Security tests for uploads, downloads, APIs and authentication.
- End-to-end tests for domestic and international transactions.
- UAT sign-off by business owners before production release.

23. Key Business Rules

Rule ID	Rule name	Required behaviour
BR-001	Product/lot separation	One product may have many lots; each lot has independent stock, quality and pricing.
BR-002	Price history	Price edits create auditable history and never silently rewrite confirmed order pricing.
BR-003	Stock reservation	Reservations are atomic and cannot exceed eligible available stock.
BR-004	Order confirmation	An RFQ/quotation is not a confirmed order until configured acceptance conditions are met.
BR-005	Price snapshot	Confirmed order items preserve agreed product, lot, quantity, price, currency and terms.
BR-006	Payment verification	Proof upload alone cannot mark funds as received.
BR-007	Shipment relation	A single order can have multiple shipments and containers; quantity allocations must reconcile.
BR-008	Document completeness	Only valid, accepted versions of required documents satisfy a checklist.
BR-009	Tracking source	Location and ETA records must identify source, timestamp and whether they are actual or estimated.
BR-010	Incoterms	Store rule, version and named place/port; do not infer legal obligations from generic statuses.
BR-011	Role authorization	Server-side permissions apply to every API and file download.
BR-012	Auditability	Critical actions record actor, timestamp, record and reason/change summary.
BR-013	Tally idempotency	Retrying a completed sync must not create duplicate accounting entries.
BR-014	Currency precision	Use fixed-precision monetary values and explicit currency codes.
BR-015	Quantity units	Quantities carry explicit units; conversions use configured rules.
BR-016	Amendments	Accepted quotations, agreements and orders use controlled amendment/version workflows.
BR-017	Cancellation	Cancellation releases eligible stock reservations and records financial/document consequences.
BR-018	Partial fulfilment	Partial shipment and delivery quantities must be traceable to the original order items.
BR-019	Data export	Exports respect record permissions and are auditable where sensitive.
BR-020	No false confirmations	UI and chatbot must not claim payment, signature, shipment or approval succeeded without authoritative confirmation.

24. Acceptance Criteria and Test Scenarios

Product and inventory

- ■ Admin can create a category, product and multiple lots for the same product.

- ■ Each lot retains its own upload timestamp, price, quality and available quantity.
- ■ Adding a new lot does not overwrite older lot data or confirmed order prices.
- ■ Two concurrent reservation attempts cannot allocate more than available stock.
- ■ Stock changes are traceable to a movement, actor and reason.

Buyer and commercial workflow

- ■ Buyer submits an RFQ with quantity, unit, quality, destination, packaging and trade terms.
- ■ Supplier can clarify requirements and issue a versioned quotation.
- ■ Buyer can accept/reject an unexpired quotation; expired quotations cannot be accepted without authorized renewal.
- ■ Order confirmation enforces required acceptance, stock and approval checks.
- ■ PDF documents show the correct buyer, supplier, lot allocation, agreed price and currency.

Payments and documents

- ■ Payment states distinguish pending, partial, verified, failed, refunded and disputed transactions.
- ■ Payment proof upload alone does not mark funds as received.
- ■ Authorized users can upload, review, approve, reject and download documents.
- ■ Superseded/rejected documents do not satisfy mandatory-document checks.
- ■ All important document revisions retain version history and actor timestamps.

Shipment and administration

- ■ Shipment milestones show expected and actual dates separately.
- ■ Tracking displays source and last update time; no unverified live location is fabricated.
- ■ Order and shipment quantities reconcile for partial deliveries.
- ■ Users cannot access unrelated orders or documents by changing an ID in a request.
- ■ Tally retries do not create duplicate entries; failed syncs are visible.
- ■ Audit logs record critical price, payment, document, order and permission changes.

24.1 Release gate

Release requires successful critical-path tests, no unresolved critical/high severity security defects, business acceptance of domestic/international workflows, verified backup restoration, production monitoring, integration failure handling and written UAT approval.

25. Phased Delivery Roadmap

Phase	Scope	Exit criteria
Phase 1 — Core marketplace	Authentication, companies, roles, categories, products, lots, pricing, inventory, RFQs and admin dashboard.	Users can publish/search lots, submit RFQs and safely manage stock.
Phase 2 — Commercial workflow	Quotations, PI, agreements, order management, payment records and order history.	A request can move from RFQ through accepted terms to a confirmed order.
Phase 3 — Documentation and logistics	Document checklists, upload/version/approval, packing lists, invoices, certificates, CHA portal, shipments, containers and milestones.	Operations can manage a shipment with traceable documents and status updates.
Phase 4 — Integrations	Tally, payment gateway, carrier tracking, GPS provider, notifications and e-signature integration as selected.	Provider tests, reconciliation, retries and audit trails pass.
Phase 5 — Automation and production readiness	Chatbot-assisted RFQ, advanced reports, performance/security testing, monitoring, backups and deployment.	UAT sign-off and production-readiness checklist completed.

For an MVP, use clearly labelled manual tracking updates and authorized payment verification where integrations are not ready. Do not label manual events as live GPS or automated provider confirmations.

26. Decisions Required Before Development

Decision ID	Question	Options / required clarification
DEC-001	Trade coverage	Domestic only, international only, or both.
DEC-002	Product publishing	Admin only, supplier self-service, or supplier submission with admin approval.
DEC-003	Lot pricing policy	Manual prices, time/date-based formula, or hybrid; define formula and approvals.
DEC-004	Stock commitment point	When is inventory reserved: quotation acceptance, order confirmation, advance payment or another event?
DEC-005	Payment setup	Bank transfer recording, gateway, partial payment, refund and international settlement requirements.
DEC-006	Tally environment	TallyPrime or other version; hosting machine; integration method; source of truth.
DEC-007	Tracking providers	Shipping lines/carriers, GPS provider, manual update roles and expected data availability.
DEC-008	Document rules	Commodity/country/route-specific mandatory and optional documents; who issues and approves each.
DEC-009	Signature provider	Required e-signature type, signer identity verification and legal review.
DEC-010	Application channels	Responsive web, Android, iOS or all; shared or separate codebase.
DEC-011	Currencies and units	Supported currency list, exchange-rate source, rounding and allowed unit conversions.
DEC-012	Approval matrix	Who approves products, prices, discounts, quotations, invoices, documents, payments and cancellations.
DEC-013	Multi-supplier sourcing	Can one RFQ be sent to multiple suppliers and can an order include multiple suppliers?
DEC-014	Shipment model	Partial shipments, multiple containers, transshipment and multiple delivery addresses.
DEC-015	Service targets	Expected user volume, performance, uptime, backup frequency, recovery objectives and retention periods.

Appendix A. Document Types and Abbreviations

Term	Meaning
B2B	Business-to-business commerce.
RFQ	Request for Quotation; buyer asks for a commercial offer.
PI	Proforma Invoice; preliminary invoice/offer document.
Packing List	Package-level details including marks, counts, contents and weights.
Tax Invoice	Invoice containing applicable tax details for the relevant jurisdiction/transaction.
Commercial Invoice	Commercial value/goods document used in trade and customs workflows where applicable.
Phytosanitary Certificate	Plant/plant-product health certificate issued by the competent authority where required.
Fumigation Certificate	Evidence of fumigation treatment where required.
COO / C/O	Certificate of Origin.
CHA	Customs House Agent; exact legal role/authorization depends on jurisdiction.
B/L	Bill of Lading; shipping document associated with carriage of goods by sea.
FOB	Free On Board; Incoterms rule whose use depends on the named port and applicable version.
CIF	Cost, Insurance and Freight; Incoterms rule for the named destination port and applicable version.
ETA	Estimated Time of Arrival; forecast, not actual arrival.
GPS	Global Positioning System; location data requires a compatible source/device.
Tally	Accounting/business management software; exact integration depends on version and environment.

Appendix B. Suggested Status Registers

B.1 Product and lot statuses

Product: Draft, Pending Approval, Active, Inactive, Archived. Lot: Draft, Pending Approval, Active, Reserved/Partially Reserved, Depleted, Blocked, Expired where applicable, Archived.

B.2 RFQ and quotation statuses

RFQ: Draft, Submitted, Under Review, Clarification Needed, Supplier Responding, Quotation Issued, Converted, Closed, Rejected. Quotation: Draft, Pending Approval, Issued, Accepted, Rejected, Expired, Superseded, Cancelled.

B.3 Order statuses

Pending Acceptance, Confirmed, On Hold, Awaiting Payment, Partially Paid, Preparing, Ready for Dispatch, In Fulfilment, Partially Shipped, Shipped, Partially Delivered, Delivered, Completed, Cancelled, Disputed.

B.4 Payment statuses

Draft, Pending, Initiated, Processing, Verification Required, Partially Paid, Paid, Failed, Rejected, Cancelled, Refunded, Partially Refunded, Disputed, Reconciled.

B.5 Document statuses

Missing, Draft, Uploaded, Submitted, Under Review, Approved, Rejected, Issued, Finalized, Expired, Superseded, Cancelled.

B.6 Shipment statuses

Planned, Booking Confirmed, Goods Preparing, Ready, Picked Up, At Origin Port, Customs Processing, Loaded, Departed, In Transit, Transshipment, At Destination Port, Customs Hold, Cleared, Out for Delivery, Delivered, Delayed, Exception, Cancelled.

B.7 Integration statuses

Not Configured, Disconnected, Healthy, Queued, Processing, Synced, Failed, Retry Scheduled, Manual Review, Reconciled.

Implementation note: This document defines proposed requirements, not a certification of legal or regulatory compliance. Before launch, validate the relevant trade rules, taxes, customs requirements, payment arrangements, certificates, privacy obligations and electronic-signature requirements with qualified business, accounting and legal professionals.